

Michal Matusik

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EDUCATION

Purdue University | West Lafayette, IN

Bachelor of Science in Computer Science | Expected May 2030

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Dart, C++, SQL

ML & Data: PyTorch, scikit-learn, XGBoost, ONNX Runtime, NumPy, pandas, OpenCV, PyArrow, SQLite

Engineering: Flutter, Next.js, React, Firebase, AWS Amplify, WebSockets, REST APIs, Git, Linux, Docker

AI-assisted development: Claude Code, MCPs, hooks, plugins

SELECTED PROJECTS

FitPerfect — On-Device Fitness Form Coach

Python, Flutter, ONNX Runtime | Apr. 2024 – May 2025

- Built a private-by-default iOS MVP that turns exercise video into local form feedback, rep counts, joint-angle analysis, and 3D motion data for squats and overhead presses.
- Hand-built the ML pipeline: YOLOv8 person detection, RTMPose keypoints, COCO-to-H36M conversion, and MotionBERT temporal 3D pose lifting; used AI-assisted development for Flutter mobile implementation.
- Engineered coordinate transforms, camera calibration, and platform-aware ONNX inference to keep skeleton overlays aligned under real capture conditions.

Chess Quant — Live Chess Tilt Detection Prototype

Next.js, TypeScript, Python, XGBoost, ONNX | May 2026 – Jun. 2026

- Built an applied-ML product that analyzes recent Lichess games and estimates when continued play may be costly, combining session-level accuracy, blunder, pace, streak, and break signals.
- Trained an XGBoost model on 1M GPU-processed games from a 90M-game corpus; retrospective simulations using the selected stopping threshold avoided 56 losses on average.
- Implemented a full-stack workflow with Lichess OAuth PKCE, Stockfish in a Web Worker, ONNX-backed inference, Firebase, Stripe, and consent-gated analytics.

Crypto Trade — Solana Token Data Acquisition Platform

Python, asyncio, WebSockets, SQLite, Parquet | May 2026 – Jul. 2026

- Developed an event-driven research platform that ingests PumpPortal migration events and collects time-stamped on-chain, market, holder, quote, risk, and alternative-data evidence.
- Designed idempotent concurrent workers, SQLite WAL lifecycle tracking, configurable risk gates, rate limits, and append-only Zstandard-compressed Parquet records for auditable downstream research.

Stockfish_v2 — Human Chess Outcome Modeling Pipeline

Python, Parquet, Stockfish 11/18 | 2026

- Built a resumable PGN-to-Parquet pipeline that joins human-game metadata with Stockfish evaluation features to support sequence-modeling of win, draw, and loss probabilities.

ACHIEVEMENT

AI Olympiad National Finalist — advanced to the national final round through competitive AI and machine-learning problem solving.